

CLASS ACTIVITY:

COURSE NAME: QUERY PROCESSING FOR DATA SCIENCE

COURSE CODE: DSA0502

NAME: Sania Herbert

REG NO:192324062

1. Missing Value Imputation

The following dataset represents students' programming marks.

STUDENT	MARKS
A	75
B	82
C	NA
D	91
E	68
F	NA
G	84

Questions:

- 1. Find the mean of the available marks.**
- 2. Replace the missing values using the mean.**
- 3. Compute the new class average after imputation.**

1. Mean of Available Marks

Available marks = 75, 82, 91, 68, 84

$$\text{Mean} = \frac{75 + 82 + 91 + 68 + 84}{5} = \frac{400}{5} = 80$$

Answer: Mean = 80

2. Replace Missing Values

C = 80

F = 80

Updated data:

Student	Marks
A	75
B	82
C	80
D	91
E	68
F	80
G	84

3. New Class Average

$$\frac{75 + 82 + 80 + 91 + 68 + 80 + 84}{7} = \frac{560}{7} = 80$$

Answer: New Average = 80

QUE: 2. Z-Score Outlier Detection

A company records the daily number of customer visits:

42, 45, 48, 50, 46, 44, 47, 49, 180

Given:

☐ **Mean = 61.22**

☐ **Standard Deviation = 42.63**

Questions:

1. Calculate the Z-score for 180.

2. Determine whether it is an outlier ($|Z| > 3$).

3. State whether the value should be removed during data cleaning.

Given

- Mean = 61.22
- Standard Deviation = 42.63
- Value = 180

1. Z-score

$$Z = \frac{180 - 61.22}{42.63} = \frac{118.78}{42.63} = 2.79$$

Answer: $Z = 2.79$

2. Is it an Outlier?

Outlier condition: $|Z| > 3$

Since $2.79 < 3$

Answer: No, it is not an outlier.

3. Remove?

Answer: No. It should not be removed.

QUE: 3. Data Normalization

A dataset contains employee salaries (₹):

42000, 56000, 70000, 85000, 98000

Questions:

1. Apply Min-Max Normalization to the salary ₹70,000.

2. Normalize all salaries into the range [0,1].

3. Identify the normalized maximum and minimum values.

Given salaries

42000, 56000, 70000, 85000, 98000

Minimum = 42000

Maximum = 98000

Formula: $\frac{x - \min}{\max - \min}$

Range

$$98000 - 42000 = 56000$$

1. Normalize ₹70,000

$$\frac{70000 - 42000}{56000} = \frac{28000}{56000} = 0.50$$

Answer: 0.50

2. Normalize All

Salary	Normalized
42000	0.00
56000	0.25
70000	0.50
85000	0.77
98000	1.00

3. Maximum & Minimum

Maximum = 1

Minimum = 0

QUE: 4. Standardization

A student's test score is 82.

Class statistics:

□ **Mean = 70**

□ **Standard Deviation = 8**

Questions:

- 1. Compute the standardized value (Z-score).**
- 2. Interpret the result.**
- 3. Would this student be considered above average?**

Given

Score = 82

Mean = 70

SD = 8

1. Z-score

$$Z = \frac{82 - 70}{8} = \frac{12}{8} = 1.5$$

Answer: 1.5

2. Interpretation

The student's score is 1.5 standard deviations above the class average.

3. Above Average?

Yes.

QUE: 5. SQL Numerical Problem

Questions:

1. Calculate the total sales amount.
2. Find the average selling price.
3. Write an SQL query to calculate total revenue.

TABLE:

Product	Quantity	Price
Laptop	8	45000
Mouse	25	650
Keyboard	15	1200
Monitor	10	14000

1. Total Sales Amount

Laptop: $8 \times 45000 = 360000$

Mouse: $25 \times 650 = 16250$

Keyboard: $15 \times 1200 = 18000$

Monitor: $10 \times 14000 = 140000$

Total

$$360000 + 16250 + 18000 + 140000 = 534250$$

Answer: ₹5,34,250

2. Average Selling Price

$$\frac{45000 + 650 + 1200 + 14000}{4} = \frac{60850}{4} = 15212.5$$

Answer: ₹15,212.50

3. SQL Query

SELECT SUM(Quantity * Price) AS Total_Revenue FROM Sales;

QUE: 6. Correlation Calculation

Given the following observations:

X 2 4 6 8 1

0

Y 3 5 7 9 11

Questions:

- 1. Compute Pearson's correlation coefficient.**
- 2. Interpret the relationship.**
- 3. Predict whether increasing X will increase Y.**

X = 2,4,6,8,10

Y = 3,5,7,9,11

1. Pearson Correlation

Since

$$Y = X + 1$$

The relationship is perfectly linear.

Answer

$$:r = 1$$

2. Interpretation

There is a perfect positive correlation.

3. Prediction

Yes. As X increases, Y also increases.

QUE:7. Grouping and Aggregation

Questions:

1. Find total sales for each region.

2. Find average sales for each region.

3. Identify the region with the highest total sales.

Region	Sales
East	250
West	320
East	280
South	300
West	350
East	270

1. Total Sales

East

$$250+280+270=800$$

West

$$320+350=670$$

South

300

Region	Total
East	800
West	670
South	300

2. Average Sales

East:

$$800/3 = 266.67$$

West: $670/2 = 335$

South

300

Region	Average
East	266.67
West	335
South	300

3. Highest Total

East (800)

QUE:8. Histogram Frequency

Frequency Questions:

1. Determine the frequency of each class.
2. Which class has the highest frequency?
3. Draw a histogram using Python.

Table

Class	Frequency
20–39	4
40–59	6
60–79	7
80–99	3

2. Highest Frequency

60–79 (7 students)

3.PYTHON CODE:

```
import matplotlib.pyplot as plt
```

```
marks = [25,30,34,36,41,45,48,52,55,57,60,63,65,67,72,75,78,82,86,90]
```

```
bins = [20,40,60,80,100]
```

```
plt.hist(marks, bins=bins, edgecolor='black')
```

```
plt.title("Histogram of Student Marks")
```

```
plt.xlabel("Marks")
```

```
plt.ylabel("Frequency")
```

```
plt.show()
```

QUE:9. Data Visualization Analysis

Questions:

- 1. Calculate the month-to-month growth.**
- 2. Determine the month with maximum increase.**
- 3. Plot the data using a line chart.**

Month	Sales
Jan	45
Feb	52
Mar	49
Apr	61
May	70
Jun	65

1. Month-to-Month Growth

From	Growth
Jan→Feb	+7
Feb→Mar	-3
Mar→Apr	+12
Apr→May	+9
May→Jun	-5

2. Maximum Increase

March → April (+12)

3.PYTHON CODE:

```
import matplotlib.pyplot as plt
months = ["Jan","Feb","Mar","Apr","May","Jun"]
sales = [45,52,49,61,70,65]
plt.plot(months, sales, marker='o')
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales (₹ Thousand)")
plt.grid(True)
plt.show()
```

10. Duplicate Detection

1. Calculate the number of unique records.
2. Compute the percentage of duplicate records.
3. Compute the percentage of missing-value records.
4. Find the total percentage of records requiring data cleaning.

Given

- Total = 2500
- Duplicate = 180
- Missing = 70

1. Unique Records

$$2500 - 180 = 2320$$

Answer: 2320

2. Duplicate Percentage

$$\frac{180}{2500} \times 100 = 7.2\%$$

3. Missing Percentage

$$\frac{70}{2500} \times 100 = 2.8\%$$

4. Total Percentage Requiring Cleaning

Total problematic records

$$180 + 70 = 250$$

$$\frac{250}{2500} \times 100 = 10\%$$

